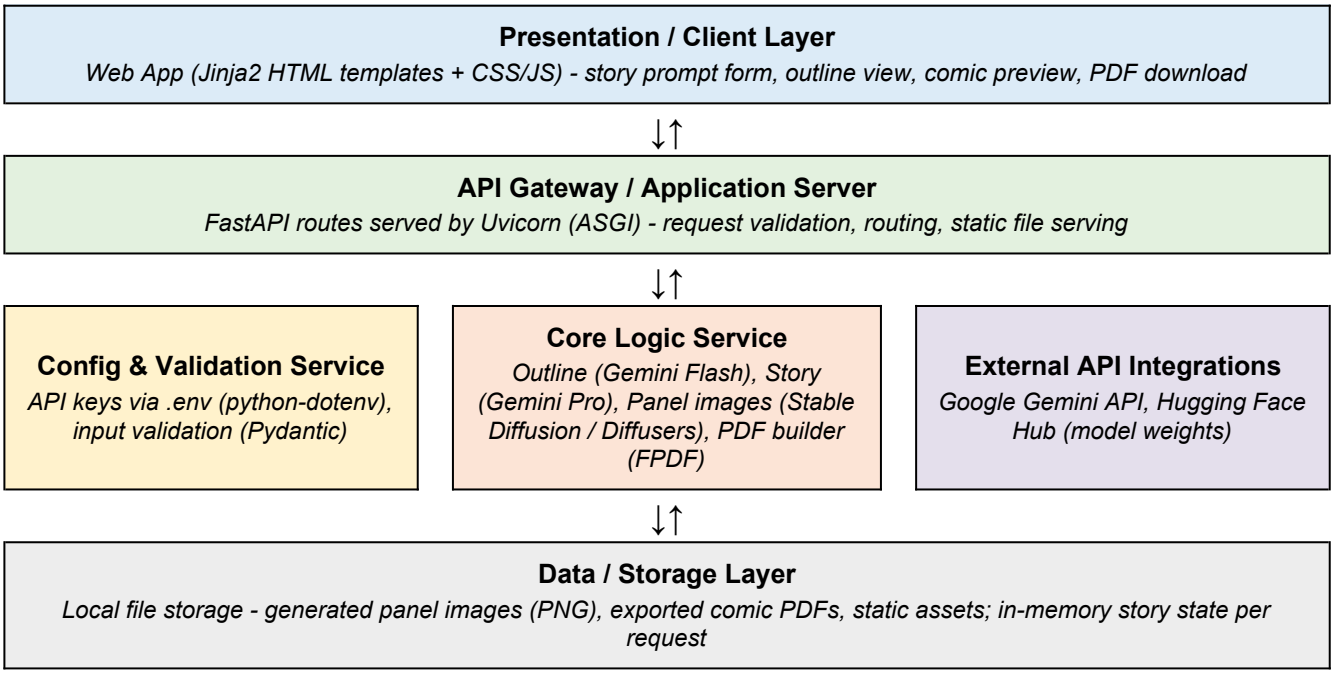
Solution Architecture

Date	3 October 2026
Team ID	SWTID-2026-5338
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Solution Architecture Diagram:

High-level view of the ComicCraft architecture, mapping the Presentation Layer, Application Layer (FastAPI gateway and AI services) and Data / Storage Layer.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Browser-based UI where the user enters a story idea, genre and number of panels, reviews the AI outline and story, views the generated comic panels and downloads the PDF.	Jinja2 templates, HTML5, CSS3, JavaScript
API Gateway	Single entry point that receives user requests, validates input, routes them to the correct service and serves generated images and PDFs back to the client.	FastAPI, Uvicorn, Pydantic, python-dotenv

Component Name	Description / Role in Architecture	Technologies Used
Microservices	Outline service (fast story skeleton), Story service (detailed scenes, dialogue and panel descriptions), Image service (one image per panel) and PDF service (compiles panels and text into a comic).	Gemini Flash, Gemini Pro (google-generativeai), Stable Diffusion with Diffusers + PyTorch, FPDF, Pillow
Database	Stores generated panel images and exported comic PDFs on disk so they can be previewed and downloaded; no training dataset is needed because pre-trained models are used.	Local file system (static/ and output/ folders), PNG and PDF files